

Fodor and Pylyshyn’s systematicity challenge still stands

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Abstract

The recent successes of neural networks producing human-like language have caused significant stir in cognitive science, with many researchers arguing that classical puzzles about human cognition and challenges to artificial intelligence are being solved by neural networks. A notable case is the argument from systematicity due to Jerry Fodor and Zenon Pylyshyn, which can be summarized as follows. Human cognition displays systematic biconditional dependencies; for example, a human can understand the sentence “John saw Mary” just in case that same human can understand the sentence “Mary saw John.” Symbolic systems account for this systematicity of language and thought immediately, due to their representing constituent structure. By contrast, neural networks can’t be guaranteed to be systematic. Several recent articles argue that this challenge has now been met by neural networks. In particular, Brenden Lake and Marco Baroni argue that their proposed *meta-learning for compositionality* protocol matches and perhaps explains human systematicity. We demonstrate that these conclusions are premature. Among other results, we found that the model displays different rates of generalization success depending on what labels are attached to what meanings. This is in sharp contrast with the fact that there are no linguistic or broader cognitive benefits from calling a carbonated beverage “pop” or “soda.” We conclude that Fodor and Pylyshyn’s challenge to neural networks remains unmet.

1 Introduction

The nineteen-eighties witnessed the maturation of a model of computation that revolutionized cognitive science and machine learning, and arguably a few decades later society at large: connectionism

(Rumelhart et al., 1986). Unlike classical symbolic models, connectionist models looked mappable to neurobiological structures, while being, in principle, capable of performing any computation classical models could. They showed great potential to reconstruct most or all of the key insights of symbolic models (Smolensky, 1988).

Three decades later, connectionist models of one form or another constitute the computational basis for the large language models that have so strongly captured our attention, both in the ivory tower and in society at large. In particular, some cognitive scientists have argued that these architectures currently provide our best theories of natural language (Piantadosi, 2024), a human capacity traditionally thought to require symbolic models as a matter of necessity. More recently, models in this class have captured public attention, and many experts and commentators consider that they hold the promise of one day, soon enough, either solving all of society’s ills or precipitating our collective demise.

Yet, already in the decade of their invention, connectionist architectures had their detractors. In a classical article from 1988, Jerry Fodor and Zenon Pylyshyn argued that this class of cognitive theories had a major and *damning* handicap compared to classical symbolic architectures (Fodor and Pylyshyn, 1988). The gist of the argument from *systematicity* is that many distinctively human faculties display biconditional interdependencies: a human has faculty *X* precisely to the extent that they have faculty *Y*. Natural language offers a crisp illustration: a human being has the ability to understand the sentence “Ann introduced Bill to Claire” precisely to the extent that that same human being can understand the sentences “Bill introduced Claire to Ann,” “Claire introduced Dan to Ed,” and so forth. Crucially, this kind of systematicity is no accident, but rather a fundamental feature of cognition that requires explanation. In par-

ticular, there aren't and there cannot be *punctate minds*, for example minds that understand "Ann introduced Bill to Claire" but are stumped by "Bill introduced Claire to Ann."

The problem, according to Fodor and Pylyshyn, is that connectionist models fail to predict, let alone explain systematicity of this kind, while classical symbolic systems predict and explain it by default.

In a classical symbolic system, the content of "Ann introduced Bill to Claire" will be represented by means of a triadic relation I which connects three terms a , b , and c , thus: $I(a, b, c)$. This representation of the complete proposition (the content of a natural language sentence or a thought) vitally *contains* the relation I , as well as each of the terms of the relation, in an explicit structural configuration which can be depicted by a tree. Thus, a symbolic system that can represent $I(a, b, c)$ can do so only because it can represent the constitutive elements of that symbolic formula in a particular structure. From this, it follows that by default such a system must be able to represent $I(b, a, c)$ and all other well-formed permutations of the symbols.¹ Additionally, a symbolic system that represents $I(a, b, c)$ will license all manner of inferences from such a representation, for example $\exists x.I(x, b, c)$ ("Someone introduced Bill to Claire"). These inferences are crucially licensed by the form of the symbolic representation itself. That is, the process that takes $I(a, b, c)$ and produces $\exists x.I(x, b, c)$ manipulates the constituents of the original representation, substituting x for a , and adding the prefix $\exists x$. Because inferences in a symbolic architecture operate on the constituents of representations, inference in these systems is, by default, systematic.

¹Why the caveat "by default"? Because a symbolic system can be contrived, of course, which blocks one or more of the possible permutations. For example, we could partition the set of all terms $T = \{a, b, c\}$ into $A = \{a\}$ and $X = \{b, c\}$, and change the relevant recursive clause from "if $t, t', t'' \in T$, then $I(t, t', t'')$ is well formed" to "if $t \in A$ and $t', t'' \in X$, then $I(t, t', t'')$ is well formed." Clearly though, this definition would be strictly more complex, it would in no way come "for free," requiring substantive motivation of one form or another. This is the sense in which classical symbolic systems are systematic by default: symbolic systems can be punctate, but they must go to significant and glaring lengths to do so. Dissatisfaction with this *argument from default* led Phillips and Wilson (2014) to propose an upgrade of the symbolic view using the formalism of category theory to pin down precisely the class of symbolic theories that truly *must* be systematic in the relevant sense.

By contrast, in a connectionist architecture the distributed network that represents "Ann introduced Bill to Claire" does not guarantee by default the possibility of representing "Bill introduced Ann to Claire." This is because a connectionist system does not by default require that "Ann," "Bill," "Claire," and "introduced" be represented as such in order to represent "Ann introduced Bill to Claire." Similarly, the inferences that a particular network may license on the basis of the representation for "Ann introduced Bill to Claire," such as "Someone introduced Bill to Claire," do not by default generalize, say to "Ann introduced someone to Claire." Since connectionist systems do not by default represent the (symbolic) constituents of (symbolically) complex representations, the processes of inference that can be defined over these systems do not, by default, involve manipulations of the constituents of complex representations.

There are excellent reasons to be critical of Fodor and Pylyshyn's broader arguments and their conclusion that connectionist systems can at best be a model of how a classical architecture is implemented in the brain. But the fragment of the argument we present here was accepted as valid and relevant by much of the field, including some of the authors' chief opponents. In particular Smolensky (1988, 1987) rejects the bleak conclusion, but wholeheartedly accepts that the systematicity that comes out essentially "for free" in symbolic architectures requires considerable ingenuity to bring out in a connectionist architecture. He proposes that a special class of connectionist systems can adequately describe and explain systematicity, in particular by means of binding units which are tensor products of role vectors and filler vectors, providing an analog of the classical notion of constituent structure.

This much will suffice in way of an exposition of the argument and the debate on systematicity. In fact, for the purpose of this article, it is just as important to clarify what the systematicity debate was *not at all* about. The systematicity debate was *not* about whether it is possible for a neural network to *behave* in a systematic fashion. Indeed, Fodor and Pylyshyn (1988) are very clear in their main recommendation for connectionists: drop your ambitions of providing a theory at the *cognitive* level, and instead see your work as a theory of the *implementation* of a cognitive-level

classical symbolic theory. Of course, for Fodor and Pylyshyn, humans behave systematically, if any organism does. If the connectionist architecture is promising as a theory of implementation of a classical symbolic cognitive theory, it follows that neural networks can, for Fodor and Pylyshyn, implement systematic behavior. Their question was whether they can properly *explain* the systematic properties of cognition, in particular by explaining what is “wrong” with punctate minds and why they do not exist.

In an article recently published in *Nature*, Brenden Lake and Marco Baroni argue that they have developed a particular protocol for meta-training neural networks which resolves this challenge from systematicity (Lake and Baroni, 2023). They focus on transformer models trained on a particular sequence-to-sequence mapping task, where strings of (non) words are mapped to strings of colored discs, and some words correspond to a disc of a particular color, while others are instructions for transforming some of the surrounding colored discs. Concretely and by way of example, the model might learn that *fep* means ● and *blicket* is an instruction to display the preceding disc three times, so that *fep blicket* would be mapped to the interpretation ●●●. After presenting the details of their meta-learning protocol and their results, we will argue that Lake and Baroni’s (2023) work fails to address, let alone solve the systematicity challenge to neural networks, for three orders of reasons.

First, their work is concerned exclusively with the *behavior* of neural networks. At no point in their article or the supplementary materials do Lake and Baroni tell us in what way their meta-learning regimen *guarantees* behavioral systematicity or *explains it* when it emerges. This was the actual nature of the challenge from systematicity and very clear to all involved in the original debate: if we grant, for argument’s sake, that a neural network can behave in *exactly* the same way as a human, what explains the systematicity of the behavior? Since, as we will see, Lake and Baroni’s work in no way changes the architecture of standard neural networks, there is simply no reason to believe, and nor do Lake and Baroni argue, that somehow the representations operative in their models have developed structures with this kind of *explanatory* power. In this regard, their work differs in a fundamental way from the ap-

proach favored at the time of the original debate, exemplified by Smolensky’s proposal in terms of tensor-product representations, whose entire *raison d’être* was to force neural networks to deal in representations which arguably encode constituent structure like symbolic models do, thereby explaining systematicity.

Second, neural networks trained with Lake and Baroni’s method fail to even *behave* systematically. We have identified three levels of behavioral failures of systematicity in these models.

- I. Within a particular grammar, they fail to assign systematic compositional meanings to a novel string that is a minor modification of a string for which the models succeed and which was reported in the article. This is analogous to understanding “Ann introduced Bill to Claire” while being mystified by “Bill introduced Claire to Ann.”
- II. Across grammars with identical rules but different words, we found that the models have different performances as a function of which words are mapped to which meanings. This behavior runs contrary to the old observation that linguistic signs are arbitrary: no benefit or handicap results from speaking a language that calls the objects of study of dendrology “trees” or “Bäume.”
- III. At the highest level of analysis, across grammars with different rules, we found that function words whose meanings differ only with respect to a constant display very different learning profiles.

Third and final, we argue that the proposed meta-learning protocol, which purports to promote systematicity and compositionality, has no plausible chance of delivering on that promise in its current form. This is because their protocol is extremely sensitive to details of the data seen during meta-learning, in a way that is in fact uncharacteristically bad at generalizing outside distribution, compared to standard meta-learning methods like model-agnostic meta-learning.

In sum, we argue that Lake and Baroni’s *meta-learning for compositionality* in no way constitutes progress towards successfully addressing the puzzle of systematicity for neural networks. Fodor and Pylyshyn’s challenge is still alive and kicking, and foundational philosophical debates from

the nineteen-eighties about the scopes of symbolic and connectionist models are as relevant today as when they were first articulated.

2 Meta-learning for compositionality

2.1 Details of the language task

To evaluate systematicity, Lake and Baroni (2023) investigate a specific sequence to sequence mapping task, where strings in simple artificial languages are mapped to their semantic interpretations. They train a standard transformer model (Vaswani et al., 2017) on this specific task using their Meta-Learning for Compositionality protocol (MLC), which we will examine in detail in section 2.2.

Each member of the family of artificial languages that they use contains *content words* denoting colored discs and *function words* denoting transformations of adjacent meanings. For example, one such language might have a content word `fep` which means $\bullet$ and a function word `blicket` which means “repeat the meaning just before three times,” so that `fep blicket` would mean $\bullet\bullet\bullet$. Table 1 shows a sample meta-learning item in detail.

All training grammars consisted of precisely four content words and precisely three function words, for example words which repeat the preceding substring n times or flip the order of two substrings. Each function word took either one or two arguments, always adjacent substrings. These arguments were always fixed to either be substrings of arbitrary complexity themselves or “atomic” content words denoting a single colored disc. Function words were only allowed to “return” (that is, to have as their corresponding meanings, once “applied” to their arguments) sequences of at least 2 discs and at most 8 discs. So “repeat the meaning of the word before 5 times” was part of the training set, but “repeat the meaning of the word before 9 times” was not. Table 2 shows examples of legal and illegal function words.

Lake and Baroni (2023) also conducted a behavioral experiment on human participants. They asked subjects to perform the same task as the models at the evaluation stage: inferring an output string from a set of input-output pairs plus a query. Human participants did this with no analog of the MLC protocol, that is no meta-learning of any kind, the natural hypothesis being that humans already think systematically. In the main

study Lake and Baroni report, all evaluations were conducted on a small subset of grammars, dubbed the “gold grammars,” which have three specific function-word meanings, and differ only with regard to which specific meanings from this common pool are assigned to which particular labels in a grammar. We describe this as a *grammar blueprint* as it defines the functional rules but not the specific color meanings or the specific words of a grammar. The three rules of the gold grammar are “after” (flip the meanings of the two substrings on either side, i.e. $x_1 f x_2 \rightarrow x_2 x_1$), “thrice” (repeat the meaning of the preceding content word three times, i.e. $u_1 f \rightarrow u_1 u_1 u_1$) and “surround” (surround the meaning of the content word to the right with two instances of the meaning of the content word to the left, i.e. $u_1 f u_2 \rightarrow u_1 u_2 u_1$).

Crucially, when learning a grammar, one content word occurs only in isolation with its meaning, never occurring in complex strings with any of the function words. Then, the query string uses one or more function words with this held-out content word, in a novel composition not seen in the evaluation set. If the model behaves systematically, it ought to correctly generalize the use of the function words on a query of novel complexity including the held-out content word.

2.2 Meta-learning for compositionality: the technique

Lake and Baroni (2023) present a novel technique they name Meta-Learning for Compositionality (MLC), which uses a standard neural network architecture to *meta-learn* over a family of tasks. We will focus here exclusively on sequence-to-sequence tasks representing language learning, since this is the example Lake and Baroni use as well, but our discussion here, like Lake and Baroni’s, extends to other kinds of tasks. Common meta-learning techniques such as Model-Agnostic Meta-Learning (MAML; Finn et al., 2017) use second-order gradients to adapt the initial parameters of a neural network by doing gradient descent on a specific task and then updating the initial parameters of a model. After meta-learning in this fashion, the model is trained further on a specific task using gradient descent, and then tested.

There is some work in cognitive science using MAML as a way of instilling in a neural network biases meant to help with generalization tasks. For example, McCoy and Griffiths (2025)

Input	Output
dax	●
fep	●
kiki	●
dax blicket	● ● ●
fep blicket	● ● ●
kiki blicket	?

Input	kiki blicket dax → ● fep → ● kiki → ●
Expected output	dax blicket → ● ● ● fep blicket → ● ● ●

(a) Examples of strings and their translation in a specific language

(b) The actual input/output pair seen by the neural network. The first string passed is the query.

Table 1: A possible training-set item with query, matching input strings in an artificial language to their interpretations as strings of colored discs, leaving the last item (the query) blank. In (a) we show a list of the input-output mappings for this item as one would most naturally use say in MAML: a list of individually processed input-output mappings, with a test item. In (b) we show how this list of mappings is collapsed into a single input-output pair in MLC: the multiple items in (a) are concatenated into a single string with a separator symbol (|), and this entire string is the input-side of a single input-output mapping, where the output is the expected response for a substring left unconnected to an output in the input (kiki blicket in this case).

use MAML to meta-train a neural network on formal languages, and demonstrate that the model, when later trained on English, shows some benefits to depth generalization.²

MLC is quite different from MAML as just summarized, see Figure 1 for a direct schematic comparison. In MAML, the meta-learning phase involves multiple cycles, each of which involves learning a different sequence-to-sequence task: several examples of input-output string mappings are presented to the network and trained using gradient descent. At the end of each cycle (known as the inner loop), the model is presented with held-out examples from the task. The model’s *initial parameters* at the beginning of the inner loop are then updated using gradient descent based on the error on the held-out examples. As the model progresses through many cycles of different tasks in the outer loop, it will settle on good initial parameters. These parameters provide the model a good start to later learn the tasks in the test dataset via gradient descent and hopefully to make the right generalizations for examples not shown in the inner loop.

In MLC, by contrast, there are no analogous

multiple cycles of meta-learning. Instead, MLC uses a single sequence-to-sequence task at the meta-learning phase, where each input sequence effectively corresponds to a cycle of MAML. That is, what in MAML was a collection of sequence-to-sequence mappings from which the network learned in the standard incremental fashion, MLC collapses into one large string (see Table 1 for a concrete example of such a collapsed string). Moreover, at the testing phase, MLC directly presents the network with an input sequence which also collapses multiple input-output mappings into one large string.

Consequently, MLC is much more limited than MAML, both at the meta-learning phase and at the test phase. At the meta-learning phase, MAML uses two orders of learning, within each cycle and then across each cycle, while MLC makes use of only one level of learning. Moreover, the one level of learning found in MLC must handle extremely complex sequences, as it were, *holistically*. This is a consequence of the strategy of taking a list of input-output mappings and collapsing it into a string to be taken as input. At the test phase, the MAML technique typically gets to benefit from yet another cycle of learning with gradient descent. The network is tested by showing it multiple examples of a new kind of input-output sequence, it gets to learn from these examples, and then it must predict the output for a given input. MLC is much more limited even at the test phase:

²Binz et al. (2023) review this and other successes of MAML in cognitive science as a means of instilling (chiefly Bayesian) biases in neural networks, while Goodale et al. (2025) challenge the interpretation of these results, arguing that MAML can be effective at instilling certain computational mechanisms (e.g. counters), rather than *specific* prior distributions.

	Schematic definition	Reason for impossibility
	$f x_1 x_2 \rightarrow x_1 x_2 x_1$	Binary functions must have their arguments surround the word.
	$f u_1 \rightarrow u_1 u_1 u_2$	Univariate functions must have arguments come before the word
	$u_1 f \rightarrow u_1$	All outputs must be at least 2 meanings long
	$u_1 f \rightarrow u_1 u_1 u_1 u_1 u_1 u_1 u_1 u_1$	All outputs must be at most 8 meanings long
Schematic definition		
$x_1 f x_2 \rightarrow x_1 x_2 x_1$		
$u_1 f \rightarrow u_1 u_1 u_1$		
$x_1 f u_1 \rightarrow u_1 x_1 u_1$		
(a) Legal functions	(b) Illegal functions	

Table 2: Examples of possible and impossible function words. Variables of the form x_i correspond to possibly complex substrings, whereas variables of the form u_i are restricted to just one (atomic) content word. Below, f is the function word, e.g. $u_1 f \rightarrow u_1 u_1 u_1$ is a function word that takes one atomic argument to its left (a single content word) and returns in interpretation a sequence of the meaning of its argument repeated three times.

instead of getting to learn from a few examples, here too it sees these examples collapsed into one long string, taken as an input. This means that, after meta-learning, the network never again adapts its parameters during testing. In other words, after meta-learning, the network is essentially tested as in a one-shot learning experiment.

In sum, Meta-Learning for Compositionality might in fact be more appositely classified as a kind of *pretraining*: what conceptually corresponds to multiple language learning tasks, each with a number of examples, gets collapsed into one large learning task on which the network is pretrained, to then be tested directly, with no additional training.

2.3 MLC in out-of-distribution tasks

It is not entirely clear how one would use MLC on a novel task which had different symbols than those encountered in training, as there is no gradient descent or other method at testing to integrate novel symbols. This means that MLC may be limited strictly to the *specific* task that the meta-training was conducted on. This greatly hinders MLC’s potential for applications, in AI as in cognitive science, since human intelligence produces systematic behavior even in entirely novel tasks, as Lake and Baroni’s (2023) own behavioral experiment attests.

Conversely, MAML can be applied to different tasks, even ones with different kinds of inputs. While the weights of an MLC model aren’t updated at test time (see the right hand side of Fig-

ure 1), with MAML it is entirely possible to add or change parts of the weights and still have them updated in gradient descent at test time. The simplest example of this is *tokenization*. Lake and Baroni’s model has a small vocabulary of nonce-words and colors, no other words or symbols can be added to the model, as there is no representation of them at test time, nor crucially a way to learn to represent new symbols. By contrast, it is very straightforward to modify the vocabulary of a MAML model to add new symbols, by expanding the embedding matrix and learning the new symbols via gradient descent at test time. Although different levels of success will be observed depending for example on the richness of the new examples, the protocol has a perfectly clear avenue to leverage gradient descent above and beyond the meta-learning phase. For example, McCoy and Griffiths (2025) use MAML to train an LSTM on various formal languages with a small, fixed vocabulary. They then train this meta-trained LSTM on an English corpus, meaning the LSTM must generalize at test time on the basis of completely new tokens. Since MLC produces a transformer with frozen weights and vocabularies, it could not perform this vital task.

One possibility, briefly considered by the authors in the discussion section, is that young children engage in many different tasks requiring systematicity, and therefore may *learn* systematic behavior by getting an analog of meta-training for compositionality with these different tasks performed very early in their lives. While the authors

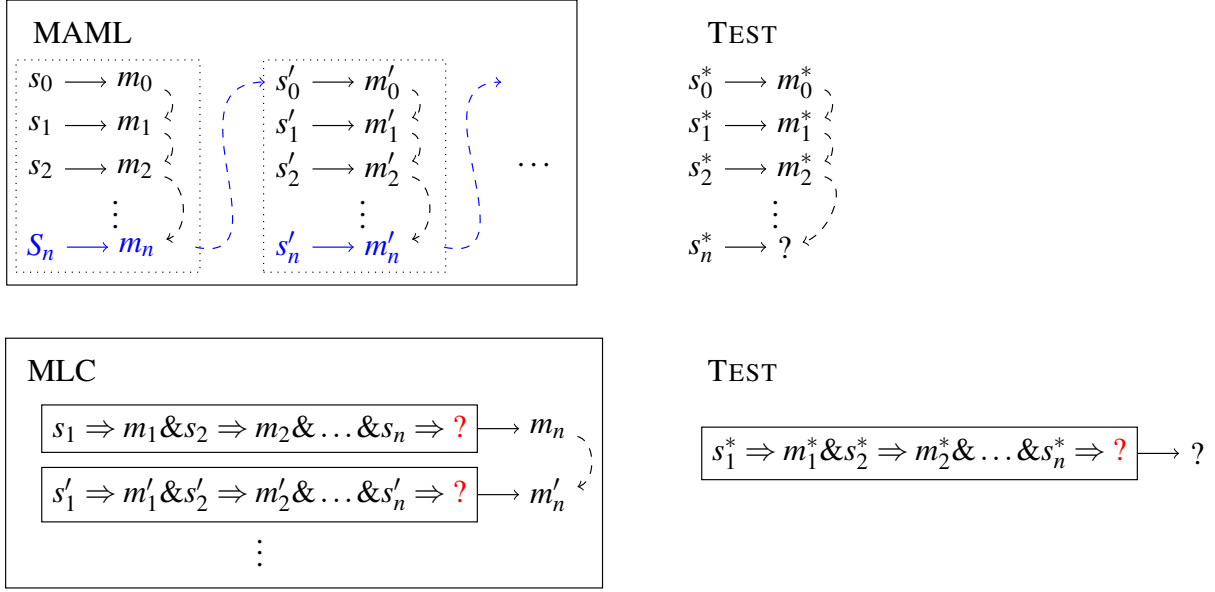


Figure 1: Schematic comparison of MAML and MLC. In MAML there are multiple cycles at meta-learning, within each of which there is incremental learning from multiple input-output mappings of strings (s_i) to meanings (m_i). At the end of each cycle the product of learning in that cycle is used to adjust the initial parameters for the next cycle. At the test phase, a new series of strings and meanings (s_i^*, m_i^*) is learned incrementally, and the model’s prediction for a novel string (s_n^*) is checked. In MLC there is only one cycle of meta-learning, where each input is a long string encoding multiple individual examples of string-to-meaning mappings, with a query (red question mark). The outputs are the correct meaning values for the missing queried element. At testing, there is no learning, a new long string encoding multiple individual examples is provided as input, and its output is checked.

do not describe how precisely this could be done with the MLC protocol, the most natural possibility is to administer MLC over an extremely diverse set of tasks. Large Language Models (LLMs) could be seen as a straightforward case study, for these models are often taken to have learned a diverse set of tasks indirectly while being trained “merely” on next-word prediction.³ Indeed, the way an MLC-trained model predicts the output given example strings bears a striking resemblance to in-context learning in LLMs. In-context learning is a phenomenon observed in LLMs where a model can perform an arguably novel task on the basis of a single or a few examples, provided as

³We hasten to add that we do not necessarily subscribe to this interpretation, and instead remain agnostic as to whether it is meaningful to say, as many do, that state-of-the-art LLMs were trained “merely” on next-word prediction yet acquired an ability to in-context learn a host of other tasks like text summation, differential medical diagnosis, psychoanalysis, or just about any task that can be operationalized in natural language. In many of our moods, we think that there is nothing “mere” about next-word prediction when the language at hand is a natural language, the unbounded representational system *par excellence*, and the training set includes just about the totality of recorded human intellectual activity.

a text prompt. For both the MLC-trained network and LLMs, when the model must infer an answer in context, it must do whatever “learning” it needs via its fixed weights, without any gradient descent. However, Lake and Baroni’s (2023) own supplementary materials pour cold water over this intriguing possibility. Despite reams and reams of pretraining arguably on a broad range of compositional tasks, GPT-4 is only able to achieve 58% accuracy on the authors’ task.

3 Failures of systematic generalization

Beyond the idiosyncrasies of the MLC approach we have described, there remains an open question as to whether MLC succeeds on its own narrowly behavioral terms. That is, under the constraints of the very particular task and evaluation set up by Lake and Baroni (2023), does the model behave systematically? To answer this question, we carefully evaluated their “gold grammar” as well as similar language blueprints with small changes to the rules. We found failures of systematicity that show that Lake and Baroni’s (2023) model fails even on their own behavior-centered criteria.

We classify these errors into three levels, in increasing (qualitative) level of abstraction. Level one failures are *within-grammar* failures, where the model succeeds at correctly interpreting a string but fails for a minor modification of it. Level two considerations look *across instantiations of a grammar blueprint* at correlations between network performance and properties of grammars which oughtn't to matter, specifically what labels (words) are mapped to what colors (meanings). At level three we look *across grammar blueprints* at the learning profiles for different rules. Our observations here concern the algebraic model, which had the best generalization performance, but these problems are present with the other variants.

To discuss the gold-grammar and our variants of it without too jarring a profusion of symbolic notation, in what follows we use c_i both to represent a specific color meaning *and* the content word for that color. Symbols like c_1 , c_2 , c_3 are the colors that were used with function words in training examples, and c_h is the held-out color used to test generalization. For example, " c_1 surround c_h " might be instantiated in a particular language as *dax kiki fep*, where *dax* would be a word seen during training as the argument of at least one function word, and *fep* would be a word seen in training only once and by itself, that is without being the argument of any function words. This string would have the schematic meaning $c_1 c_h c_1$, as the function word, *kiki*, means "surround."

3.1 Level one failures

Lake and Baroni report several successfully generalized strings, but we found the model fails for strings of comparable complexity. A striking example is the schema " c_1 surround c_h after c_h thrice," which is incorrectly predicted in many grammars (see Table 3 and the supplementary materials). Importantly, this schema is a very minor variation of one tested in the article and successfully learned within the same grammars that fail our variant: " c_h surround c_h after c_h thrice." The two schemata are different special cases of " x surround y after z thrice," showing that performance differs dramatically between equally valid combinations of x , y , z (c_h , c_h , c_h vs. c_1 , c_h , c_h). This is akin to a human understanding the English sentence "Ann introduced Bill to Claire" while being stumped by "Dan introduced Claire to Anne," betraying a deep failure of systematicity in its most

paradigmatic manifestation.

3.2 Level two failures

The model is sensitive to precisely which labels (words) are mapped to which meanings (rules and colors), with different mappings producing different performances. Thus, the networks do not conform to the arbitrariness of linguistic signs, whereby no particular benefits or drawbacks are created by American English "sofa" vs. (Old) Canadian English "chesterfield."

For example, the same schema discussed above (" c_1 surround c_h after c_h thrice") shows different learning successes *across* grammars, being correctly generalized for some label-color pairs and not for others (Table 3). Additionally, we found that the model has different success rates on the schema, depending on precisely which color is held out (Table 4).

3.3 Level three failures

We found that the network is overly sensitive to what it saw at the meta-learning phase. The meta-learning protocol only exposed the networks to rules whose outputs had lengths between 2 and 8 words, and we found that they could not learn a rule that was a trivial extension, violating this 2–8 constraint. Specifically, while the model can learn a function that takes one argument and repeats it n times for $2 \leq n \leq 8$, it couldn't learn an analogous rule that repeats its argument, say, 9 times. This cannot be due to traditional constraints on production such as limited memory, as the model was unable to learn to repeat its argument *once*, showing it failed to learn even the identity function. This shows that the model isn't a *systematic learner*, instead, it is entirely beholden to what precisely appeared in the meta-learning phase. Additionally, we found that performance within the 2–8 length bounds drops as the output gets longer, mirroring the frequency with which functions were encountered at meta-learning (see Figure 2)

Perhaps most damningly, the model fails to generalise to new lengths even on a simple copying task. Copying requires providing the output of an input string which is in the prompt as an input-output pair. In other words, it requires nothing more than the ability to copy a string and yet the model is still incapable of doing it when provided any rule outside the 2–8 length range.

Output	Count across label-color pairs	
	Static order	Shuffled order
$c_h c_h c_h c_1 c_h c_1$	1993	2785
$c_1 c_h c_h c_h c_h c_1$	1045	934
$c_h c_h c_h c_h c_1 c_1$	1282	598
$c_h c_h c_h c_h c_1 c_h$	0	3

Table 3: Four very different outputs for the same schema “ c_1 surround c_h after c_h thrice,” with counts across label-color pairings. The first is the normatively correct output. The second is a possible output if one parses the string as “ x surround (y after z thrice),” though technically this was unintended with the gold grammar, as “surround” examples only ever take two primitives (color-denoting labels) as arguments. The third and fourth aren’t acceptable outputs under any parse. *Static order* vs. *shuffled order* concerns the order in which the study items were presented to the network. With static order of presentation the network’s accuracy on this schema is at most 70%, with shuffled order at most 86%. The different results depending on order of presentation are instructive, as they indicate their own kind of non-systematic fragility.

Held out color	Constant presentation order		Shuffled presentation order	
	% correct	SEM	% correct	SEM
Blue	56.11	1.8506	59.72	1.8291
Green	62.08	1.8094	73.47	1.6464
Pink	38.47	1.8144	67.36	1.7487
Purple	50.28	1.8647	69.03	1.7244
Red	53.75	1.8594	75.83	1.5965
Yellow	16.11	1.3710	41.39	1.8368

Table 4: Accuracy on held-out-label-color combinations for the string “ c_1 surround c_h after c_h thrice.”⁴

4 Discussion and conclusions

We found three different levels of failures of systematic generalization in Lake and Baroni’s algebraic network. We conclude that, while the network did show some measure of success at later learning on the basis of extremely small datasets, these successes cannot be described as instances of systematic generalization, so that the systematicity challenge is in fact *not* addressed by the MLC protocol, in its current shape.

One counter to our conclusion, consonant with the methodology of the article which included behavioral data from humans, is to propose that neural networks need only be as systematic as humans are *in performance*. There are all manner of reasons why a particular human in a particular circumstance might behave differently on the basis of the two stimuli “Ann introduced Bill to Claire”

and “Claire introduced Dan to Ed.” Having had an unpleasant experience with someone named Bill might do the trick. Cognitive science recognizes that a faculty may be systematic, while particular circumstances of use of a faculty might not show it.

Now, we strongly suspect that the failures of systematicity we presented here would not be observed in humans. Readers might not share our expectation, but we submit that our findings pose an important challenge to the conclusions of the target article nonetheless. Because the puzzle the article ostensibly attempts to solve concerns *competence*, one of two positions would have to be argued for.

The first option is to accept the classical points about systematicity, despite performance complications, and to somehow investigate the *competence* of MLC networks, to show that *it* is systematic in the required sense (Firestone, 2020). At points, the authors imply that what they call the algebraic model is a model of competence, while

⁴We thank Brenden Lake for suggesting that we consider the performance of the model if the data are presented shuffled each time.

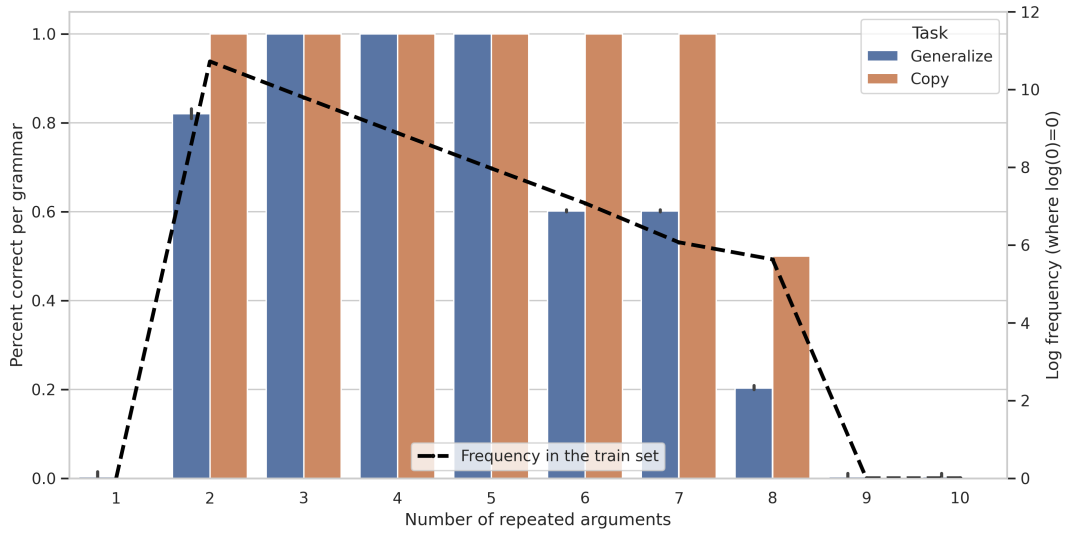


Figure 2: Learning functions that take an argument x and return n repetitions of x . *Generalize* strings are the original test sequences where we replace thrice by n . *Copy* shows the model’s ability to repeat sequences that are provided as study examples (i.e. lookup). The figure shows confidence intervals but the distribution has so little variation that they are hard to see.

the models supplemented with heuristics and human performance data would be models of performance. Yet, our discoveries here concern precisely the algebraic model, and show that, at least on its face, it is not systematic. This means that anyone pursuing this direction will have their work cut out for them, as they will have to find some aspect of the internal structure of these models that somehow constitutes a capacity for perfectly systematic powers of generalization. And to our knowledge, no current framework for mechanistic interpretation of neural networks provides methods to find *capacities* in the sense familiar from cognitive science as would be required here, crucially distinct from the *performance* of the networks at actual testing.

The second option is to reject the classical arguments: perhaps classical systematicity is actually *not* a property of human cognition, or competence is too idealized a notion to be of real interest, or any number of other responses that argue Fodor and Pylyshyn’s challenge away. Since we cannot rehearse here the full systematicity debates from the articles we cite in the introduction, all we can do is urge the reader to (re)visit this wonderful scholarship and the extremely productive and enlightening debate it inspired, in order to decide for themselves.

What is clear, however much weight one is inclined to put on Fodor and Pylyshyn’s argument from systematicity, is that their challenge still stands, and that meta-learning (for compositionality or otherwise), while interesting and potentially extremely useful, has yet to fulfill its promise to bridge this particularly important gap between symbolic and connectionist approaches. Current neural networks with generic architectures, that is models without hard and strong prior constraints tailored to guarantee emulation of the core properties of symbolic systems, neither produce systematic behavior nor explain how systematicity can be a fundamental property of minds.

Author contribution statements

Both authors were involved in the study’s conception and design. Programming and running experiments was handled by MG. Both authors contributed to the writing and editing of the article.

Code availability

All relevant code for these experiments, allowing for full reproduction of our results, can be found at <https://github.com/MichaelGoodale/mlc-reply>.

Data availability

All relevant data is included in the repository of our code or can be re-created by running our code.

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